Rahul Khatri

EV Systems Engineer

RESUME_1118

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Total Experience: 6 years

PROFESSIONAL SUMMARY

EV Systems Engineer with 6 years of experience in ev systems, using Battery Management Systems, CAN Bus, MATLAB, Simulink. Experienced in model-based design, automotive safety and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: HIL, Simulink, Battery Management Systems, Motor Control, AUTOSAR, SOC Estimation, MATLAB, ISO 26262, Diagnostics, EV Powertrain, CAN Bus, Python

Methods: Model-based design, Automotive safety, Vehicle networks

PROFESSIONAL EXPERIENCE

Senior EV Systems Engineer | JuniperWorks Systems | Jan 2023 - Present

- Lowered peak powertrain losses by 41% in control-model experiments
- Reduced simulated fault-detection delay by 30% using diagnostic timing refinements
- Validated 551 HIL fault-injection scenarios for protective responses
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate EV Systems Engineer | Foresight Industrial | Jul 2020 - Dec 2022

- Validated 832 HIL fault-injection scenarios for protective responses
- Improved SOC estimation error by 36% after parameter calibration

PROJECTS

Electric Drive Controller | Simulink, CAN Bus, SOC Estimation, EV Powertrain

- Modeled torque request logic and motor protection interlocks.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

Charging Control Study | CAN Bus, Python, AUTOSAR, Simulink

- Simulated voltage and current control under load variations.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

EDUCATION

B.Tech in Electrical Engineering, Pragati Institute of Applied Sciences, 2020
Relevant Methods: Model-based design, Automotive safety, Vehicle networks

CERTIFICATIONS

Functional Safety Basics (2022)